

Methods 3: Multilevel Statistical Modeling and Machine Learning

Week 01: *Introduction*
September 11, 2026

Outline

- 1) What you already know (Methods 1&2)
- 2) Motivation for multilevel modelling
- 3) Motivation for machine learning
- 4) Presenting the instructor and myself
- 5) Academic regulations and exam

RATIONALE OF METHODS COURSES

Providing you flexible tools for
estimating models

1) What you already know (Methods 1&2)

The foundations we build upon

[← COURSE CATALOGUE](#)

Methods 1: Introduction to Experimental Methods, Statistics, and Programming

Autumn semester 2023

The course includes an introduction to experimental methods (including experimental design and data collection methods), statistics, and basic programming (using for example R).

Student's Sleep Data

Description

Data which show the effect of two soporific drugs (increase in hours of sleep compared to control) on 10 patients.

```
data(sleep)
print(sleep)
```

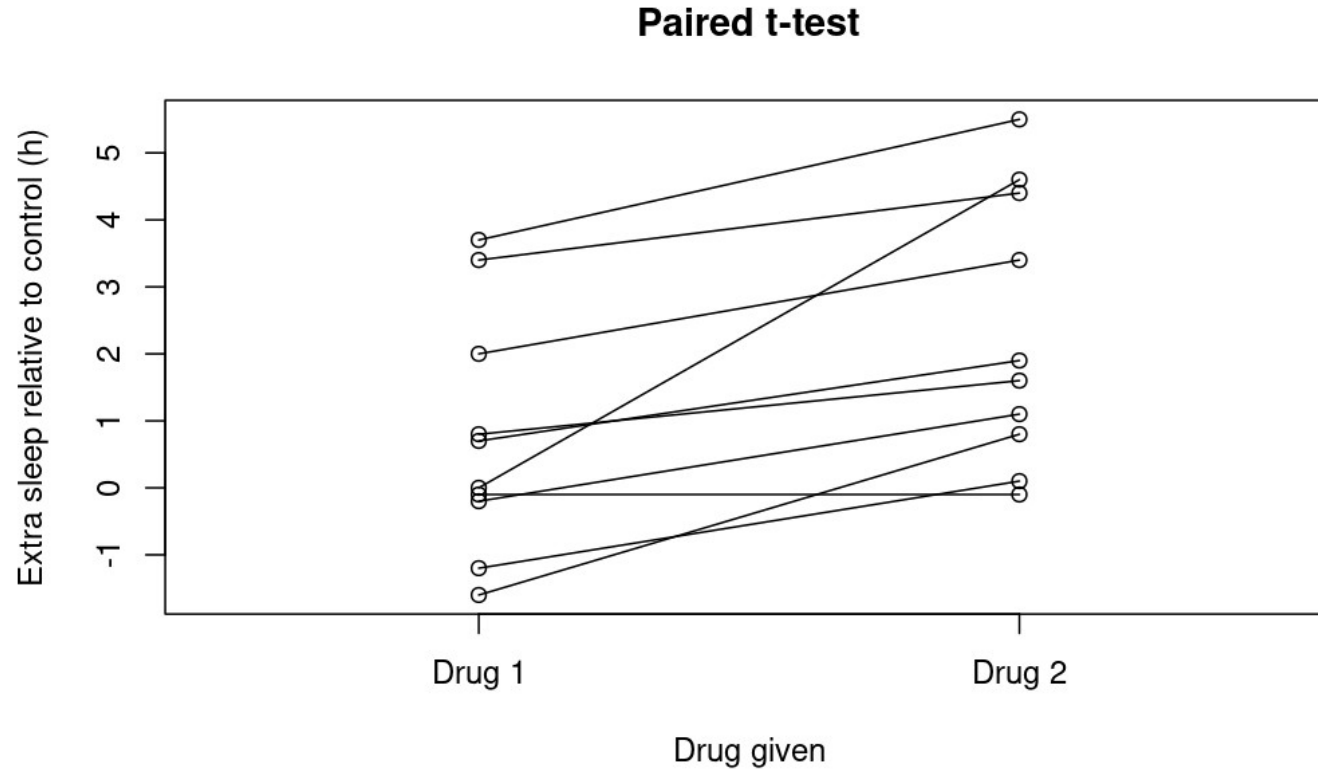
```
##      extra group ID
## 1      0.7      1  1
## 2     -1.6      1  2
## 3     -0.2      1  3
## 4     -1.2      1  4
## 5     -0.1      1  5
## 6      3.4      1  6
## 7      3.7      1  7
## 8      0.8      1  8
## 9      0.0      1  9
## 10     2.0      1 10
## 11     1.9      2  1
## 12     0.8      2  2
## 13     1.1      2  3
## 14     0.1      2  4
## 15    -0.1      2  5
## 16     4.4      2  6
## 17     5.5      2  7
## 18     1.6      2  8
## 19     4.6      2  9
## 20     3.4      2 10
```

The foundations we build upon

```
print(t.test(sleep$extra[sleep$group == 1], sleep$extra[sleep$group == 2],  
            paired = TRUE))
```

```
##  
## Paired t-test  
##  
## data:  sleep$extra[sleep$group == 1] and sleep$extra[sleep$group == 2]  
## t = -4.0621, df = 9, p-value = 0.002833  
## alternative hypothesis: true difference in means is not equal to 0  
## 95 percent confidence interval:  
##  -2.4598858 -0.7001142  
## sample estimates:  
## mean of the differences  
## -1.58
```

The foundations we build upon



Your Swiss Army Knife

After Methods 1



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The foundations we build upon

← COURSE CATALOGUE

Methods 2: The General Linear Model

Spring semester 2024

The course includes instruction on the General Linear Model. The course includes theoretical lectures on the GLM and practical exercises in programming.

This course builds on the Methods 1 course and provides the deeper knowledge of statistics required to acquire more advanced statistical methods.

The course prepares students for multilevel modelling, machine learning, and Bayesian computational modelling in the Methods 3 and Methods 4 courses.

The General Linear Model is an
extremely flexible tool

```
data(mtcars)
print(mtcars)
```

RStudio: Notebook Output

	mpg <dbl>	cyl <dbl>	disp <dbl>	hp <dbl>	drat <dbl>	wt <dbl>
Mazda RX4	21.0	6	160.0	110	3.90	2.620
Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875
Datsun 710	22.8	4	108.0	93	3.85	2.320
Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215
Hornet Sportabout	18.7	8	360.0	175	3.15	3.440
Valiant	18.1	6	225.0	105	2.76	3.460
Duster 360	14.3	8	360.0	245	3.21	3.570
Merc 240D	24.4	4	146.7	62	3.69	3.190
Merc 230	22.8	4	140.8	95	3.92	3.150
Merc 280	19.2	6	167.6	123	3.92	3.440
Merc 280C	17.8	6	167.6	123	3.92	3.440
Merc 450SE	16.4	8	275.8	180	3.07	4.070
Merc 450SL	17.3	8	275.8	180	3.07	3.730
Merc 450SLC	15.2	8	275.8	180	3.07	3.780
Cadillac Fleetwood	10.4	8	472.0	205	2.93	5.250

Description

The data was extracted from the 1974 *Motor Trend* US magazine, and comprises fuel consumption and 10 aspects of automobile design and performance for 32 automobiles (1973–74 models).

Format

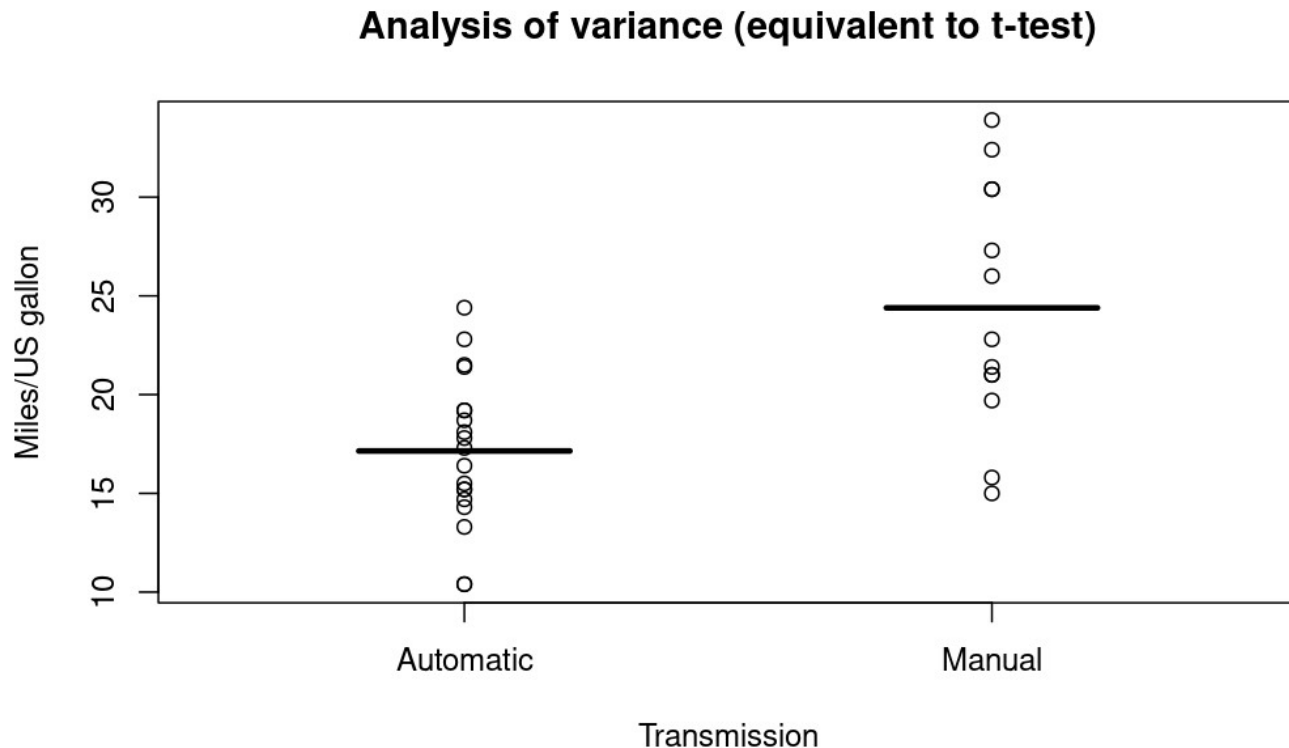
A data frame with 32 observations on 11 (numeric) variables.

[, 1] mpg Miles/(US) gallon
[, 2] cyl Number of cylinders
[, 3] disp Displacement (cu.in.)
[, 4] hp Gross horsepower
[, 5] drat Rear axle ratio
[, 6] wt Weight (1000 lbs)
[, 7] qsec 1/4 mile time
[, 8] vs Engine (0 = V-shaped, 1 = straight)
[, 9] am Transmission (0 = automatic, 1 = manual)
[,10] gear Number of forward gears
[,11] carb Number of carburetors

Call:
lm(formula = mpg ~ factor(am),
data = mtcars)
Coefficients:
(Intercept) factor(am)1
17.147 7.245

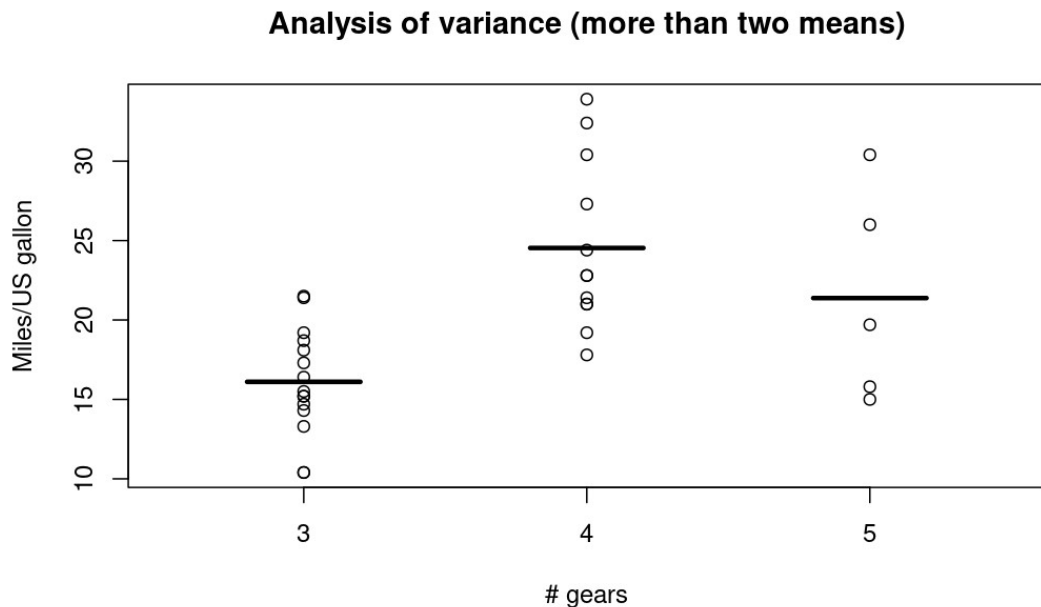
```
Call:
lm(formula = mpg ~ factor(am), data = mtcars)
Coefficients:
(Intercept)  factor(am)1
    17.147         7.245
```

*We can do the
same as in
Methods 1*



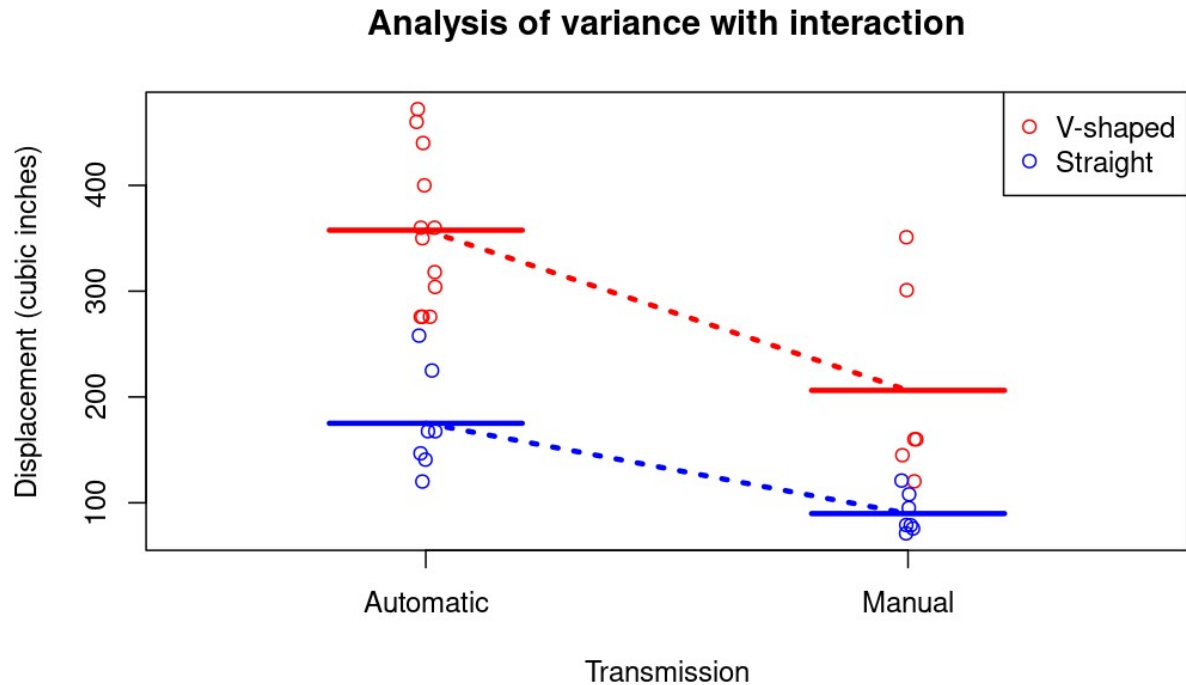
```
Call:
lm(formula = mpg ~ factor(gear), data = mtcars)
Coefficients:
 (Intercept)  factor(gear)4  factor(gear)5
      16.107         8.427         5.273
```

*... but we can
extend to factors
(gear) with more
than two levels (3,
4 or gears)*



```
Call:
lm(formula = disp ~ factor(vs) * factor(am), data = mtcars)
Coefficients:
      (Intercept)          factor(vs)1          factor(am)1  factor(vs)1:factor(am)1
            357.62             -182.50             -151.40              66.09
```

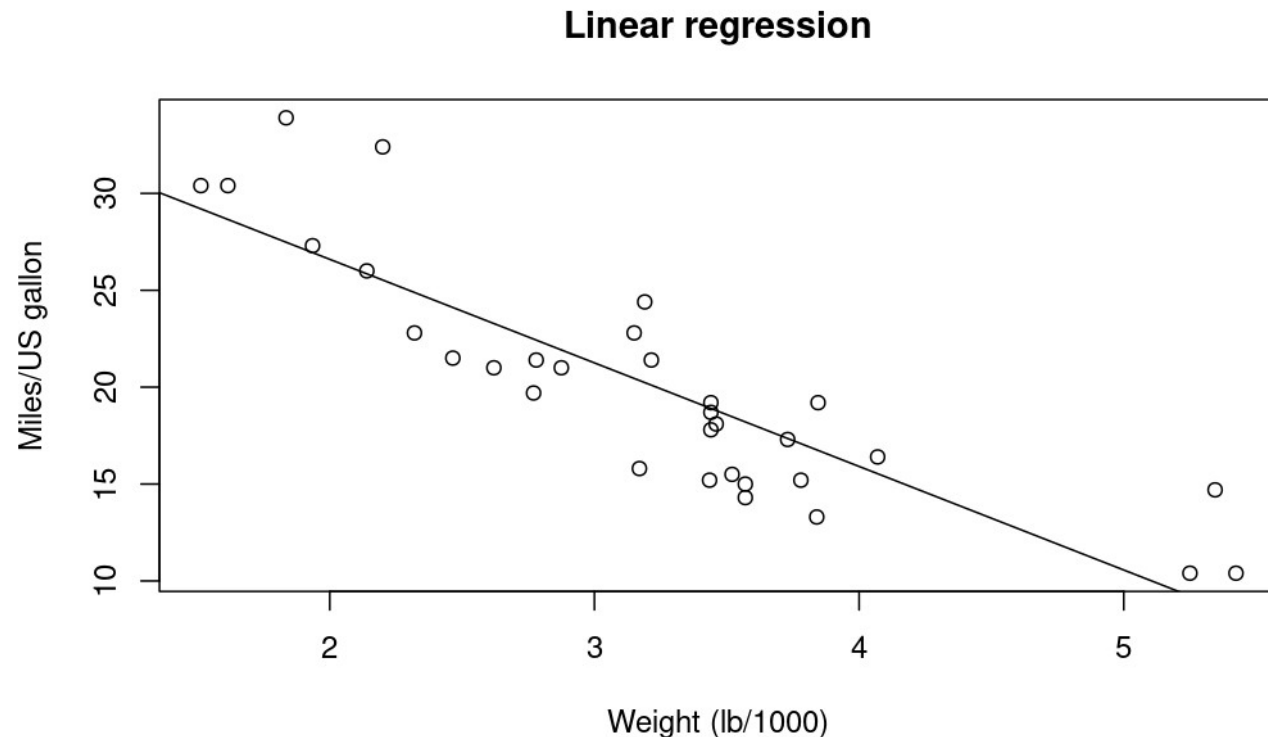
... and we can model interactions, (i.e. one independent variable (vs) depends on another independent variable (am))




```
Call:
lm(formula = mpg ~ wt, data = mtcars)
Coefficients:
(Intercept)          wt
      37.285       -5.344
```

*... and we can
subsume linear
regression under the
general linear model*

$$y = ax + b$$



Call:

```
lm(formula = mpg ~ I(wt^2) + wt, data = mtcars)
```

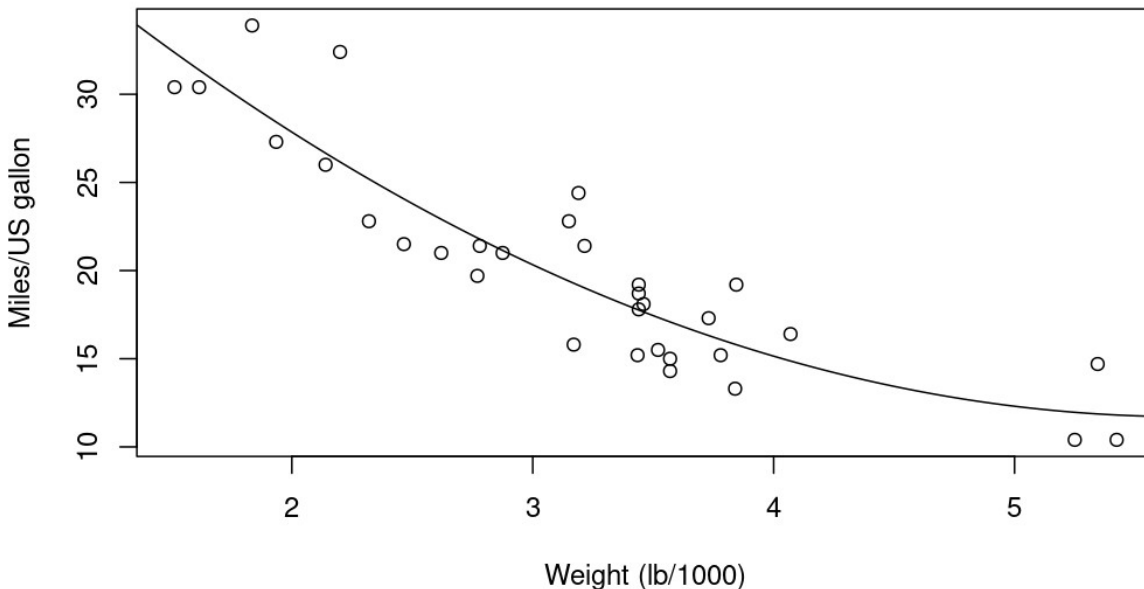
Coefficients:

(Intercept)	I(wt^2)	wt
49.931	1.171	-13.380

Quadratic regression

*... and quadratic
regression*

$$y = ax^2 + bx + c$$

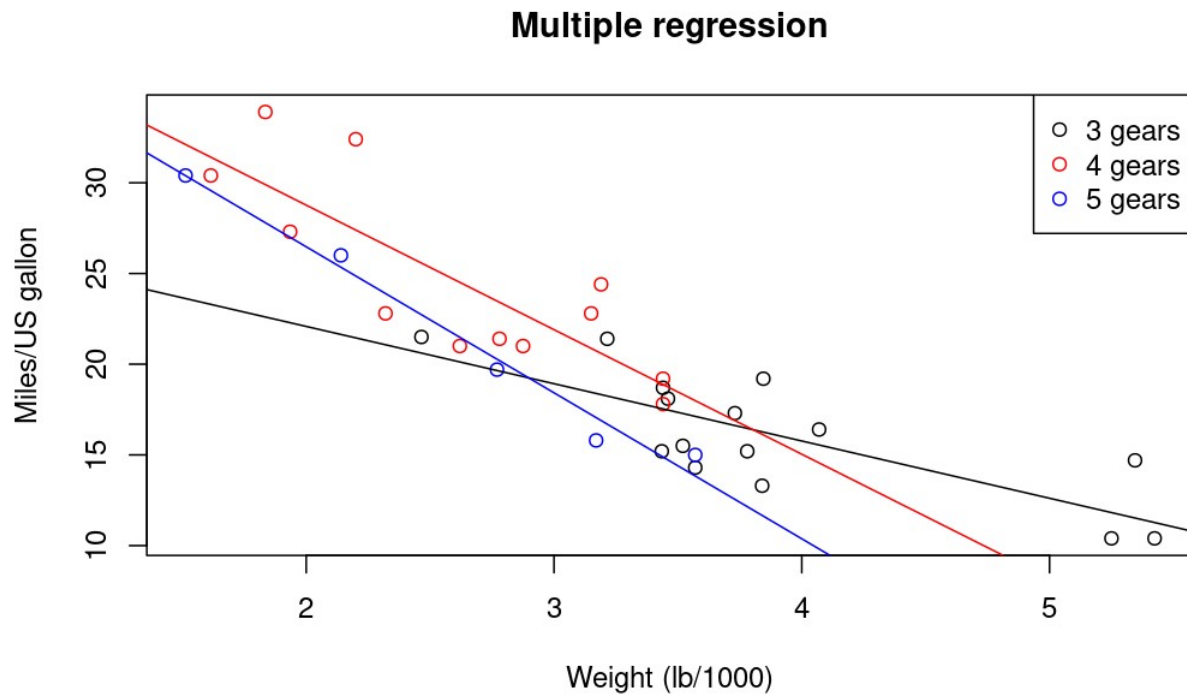


Call:
`lm(formula = mpg ~ wt * factor(gear), data = mtcars)`

Coefficients:

	wt	factor(gear)4	factor(gear)5	wt:factor(gear)4	wt:factor(gear)5	
(Intercept)	28.395	-3.157	14.098	14.168	-3.707	-4.889

... and multiple regression



The beauty of the General Linear Model

$$Y = X\beta + \epsilon$$

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

Y : a column vector with n observations (known)

X : the design matrix (known), size: $n \times p$

β : a column vector with p (unknown) model parameters

ϵ : a column vector with n residuals, $N(0, \sigma^2)$

The design matrix: X

what is this?

what is this?

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \\ y_7 \end{bmatrix} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \\ 1 & x_4 \\ 1 & x_5 \\ 1 & x_6 \\ 1 & x_7 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \varepsilon_4 \\ \varepsilon_5 \\ \varepsilon_6 \\ \varepsilon_7 \end{bmatrix}$$

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \\ y_7 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \varepsilon_4 \\ \varepsilon_5 \\ \varepsilon_6 \\ \varepsilon_7 \end{bmatrix}$$

$$Y = X\beta + \epsilon$$

Your Swiss Army Knife

After Methods 2



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Your Swiss Army Knife

After Methods 3



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General multilevel linear model

$$Y = X\beta + Zu + \epsilon$$

Y : a column vector with n observations (known)

X : the first level design matrix (known), size: $n \times p$

β : unknown $p \times 1$ column vector of the first level coefficients

Z : $n \times q$ design matrix (known) for the q random effects

u : unknown $q \times 1$ column vector of the second-level coefficients: $N(0, G)$

G : the covariance matrix of the second-level effects (unknown): $q \times q$

ϵ : $N \times 1$ column vector (unknown) of the residuals: $N(0, \sigma^2 I)$

General multilevel linear model

$$Y = X\beta + Zu + \epsilon$$

Now we have two design matrices,
 X and Z , but no analytical solution.
(What does it mean and entail that
we have no analytical solutions?)

General multilevel linear model

$$Y = X\beta + Zu + \epsilon$$

Henderson's equation

$$\begin{pmatrix} X^T X & X^T Z \\ Z^T Z & Z^T Z G^{-1} \end{pmatrix} \begin{pmatrix} \hat{\beta} \\ \hat{u} \end{pmatrix} = \begin{pmatrix} X^T y \\ Z^T y \end{pmatrix}$$

General multilevel linear model

$$Y = X\beta + Zu + \epsilon$$

Henderson's equation

$$\begin{pmatrix} X^T X & X^T Z \\ Z^T Z & Z^T Z G^{-1} \end{pmatrix} \begin{pmatrix} \hat{\beta} \\ \hat{u} \end{pmatrix} = \begin{pmatrix} X^T y \\ Z^T y \end{pmatrix}$$

Maximise this

$$l_{ML}(\hat{\beta}, \hat{u}, \hat{\sigma}^2 | y) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \log |G| - \frac{1}{2} \log |G^{-1} + Z^T Z| - \frac{n}{2} \log\left(\frac{r^2}{n}\right) - \frac{n}{2}$$

Your Swiss Army Knife

After Methods 4



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Why is the Swiss Army Knife an apt metaphor?

- Each tool is independent of everything else and works perfectly fine on its own
- Sometimes all you need is a blade (t-test)
- Flexibility comes at a cost (the Methods 4 knife is heavier to wield)

A misconception

Paraphrase of what I have overheard in
conversations:

“In Methods 4 we are going to learn the correct way, the other courses represent the wrong way”

2) Motivation for multilevel modelling

The foundation we are building

← COURSE CATALOGUE

Methods 3: Multilevel Statistical Modeling and Machine Learning

Autumn semester 2021

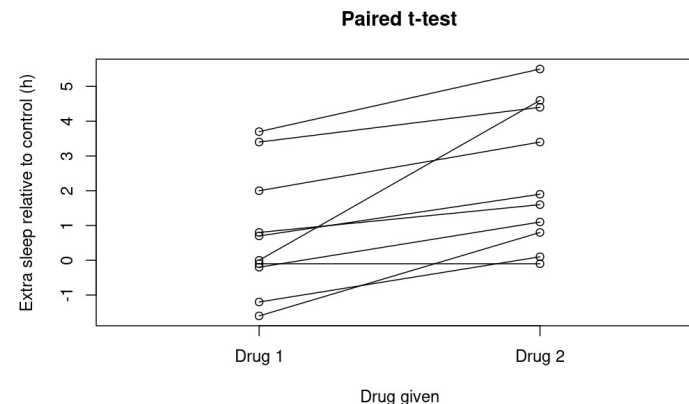
The course includes instruction on statistical techniques relying on the Generalised Linear Model, including multilevel modelling and basic machine learning procedures (e.g. predictive approaches and validation techniques).

The course includes lectures on statistical and machine learning concepts as well as practical exercises in programming.

When might the General Linear Model
allow for too little flexibility?

Advantages of multilevel modelling

Paired t-test
data: sleep\$extra[sleep\$group == 1] and sleep\$extra[sleep\$group == 2]
t = -4.0621, df = 9, p-value = 0.002833
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
-2.4598858 -0.7001142
sample estimates:
mean of the differences
-1.58



Call:
lm(formula = extra ~ factor(group), data =
sleep)

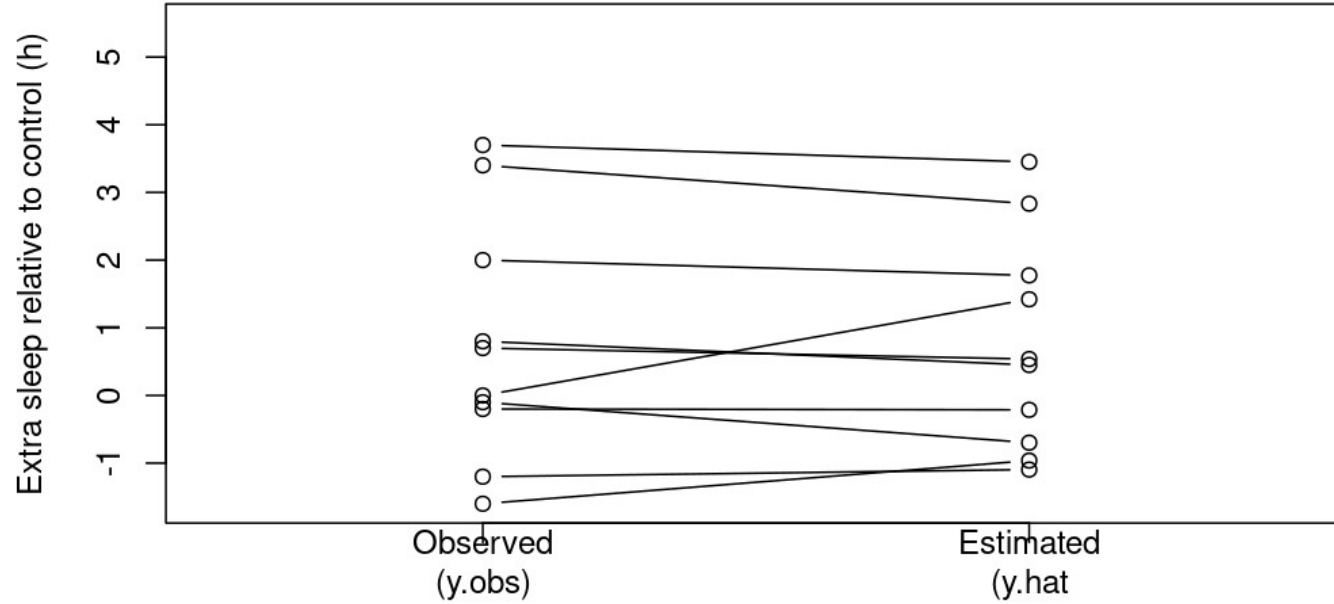
Coefficients:
(Intercept) factor(group)2
0.75 1.58

Can explain more variance

Linear mixed model fit by REML ['lmerMod']
Formula: extra ~ factor(group) + (1 | ID)
Data: sleep
REML criterion at convergence: 69.9559
Random effects:
Groups Name Std.Dev.
ID (Intercept) 1.6877
Residual 0.8697
Number of obs: 20, groups: ID, 10
Fixed Effects:
(Intercept) factor(group)2
0.75 1.58

```
> sigma(lm(extra ~ factor(group), data=sleep))
[1] 1.898625
```

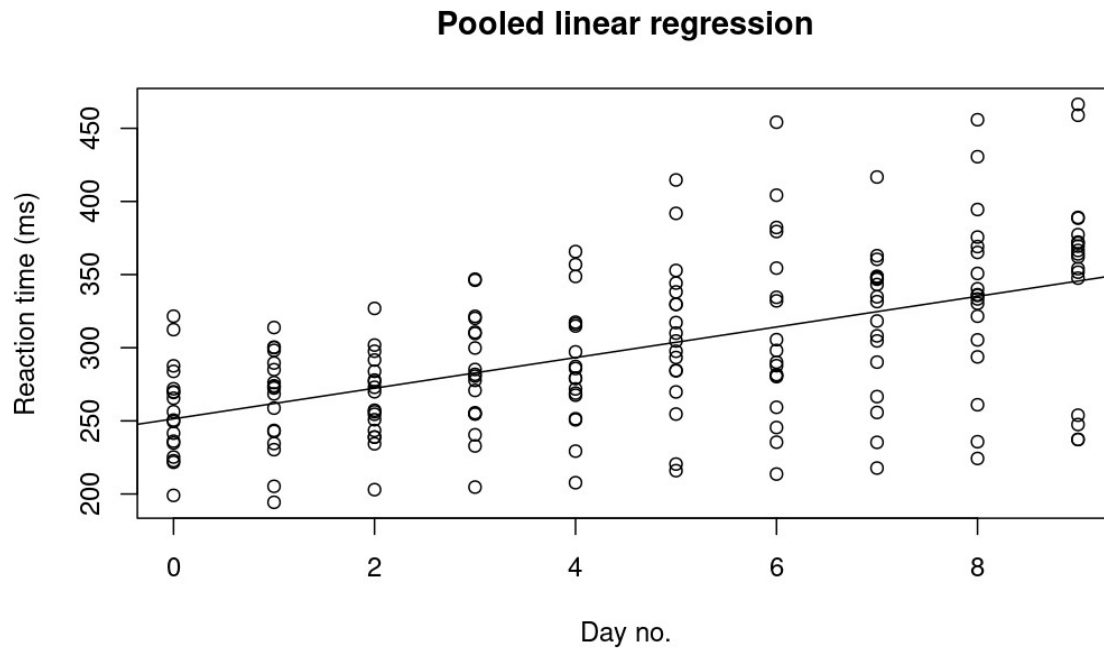
Observed vs. estimated effect of Drug 1



*Can
estimate
single
subjects*

```
> print(str(sleepstudy))
'data.frame':  180 obs. of  3 variables:
 $ Reaction: num  250 259 251 321 357 ...
 $ Days     : num   0  1  2  3  4  5  6  7  8  9 ...
 $ Subject  : Factor w/ 18 levels "308","309","310",...: 1 1 1 1 1 1 1 1 1 1 ...
```

```
Call:
lm(formula = Reaction ~ Days, data = sleepstudy)
Coefficients:
(Intercept)      Days
    251.41         10.47
```



Linear mixed model fit by REML ['lmerMod']
 Formula: Reaction ~ Days + (1 | Subject)
 Data: sleepstudy
 REML criterion at convergence: 1786.465
 Random effects:

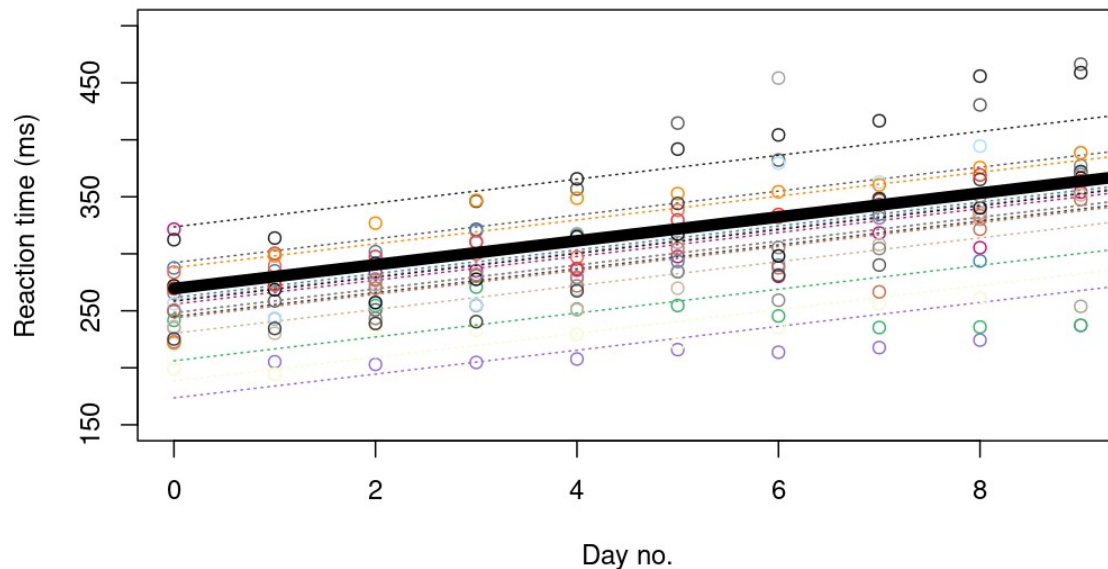
Groups	Name	Std.Dev.
Subject	(Intercept)	37.12
Residual		30.99

 Number of obs: 180, groups: Subject, 18
 Fixed Effects:

	Days
(Intercept)	251.41
	10.47

Respects data structure

Linear regression with subject-level intercepts



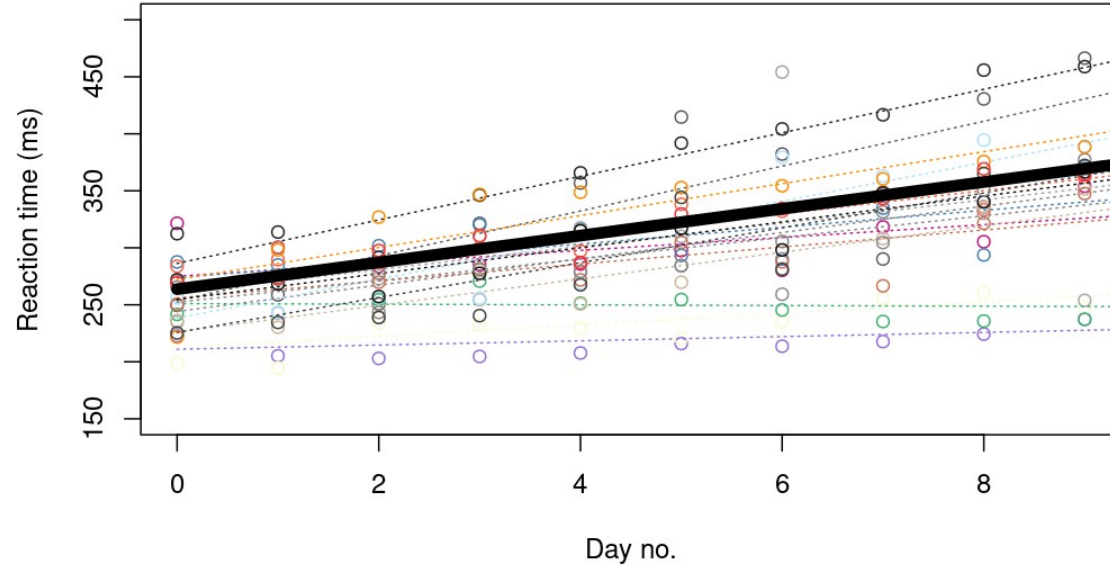
```

Linear mixed model fit by REML ['lmerMod']
Formula: Reaction ~ Days + (Days | Subject)
Data: sleepstudy
REML criterion at convergence: 1743.628
Random effects:
Groups   Name             Std.Dev. Corr
Subject  (Intercept) 24.741
          Days      5.922   0.07
Residual                25.592
Number of obs: 180, groups: Subject, 18
Fixed Effects:
(Intercept)           Days
    251.41         10.47

```

Respects data structure

Linear regression with subject-level intercepts and slopes



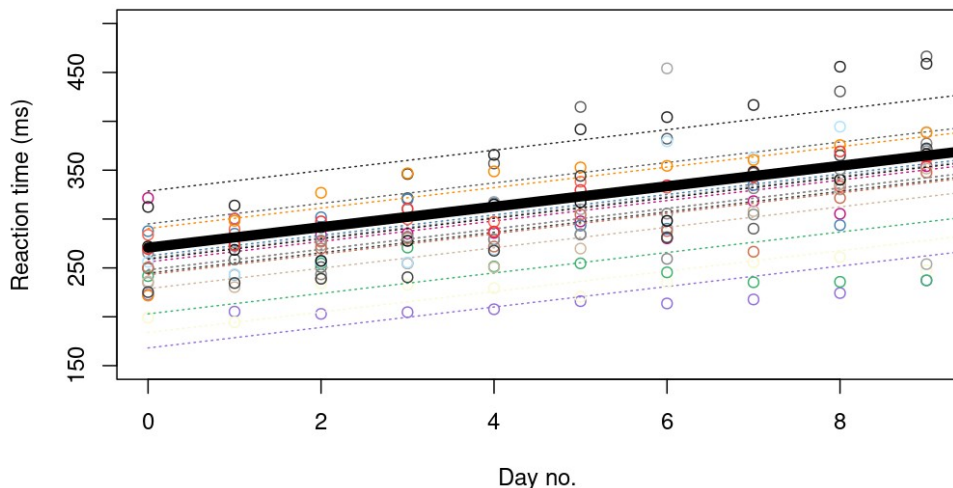
Call:

```
lm(formula = Reaction ~ Days + Subject - 1, data = sleepstudy)
```

Coefficients:

Days	Subject308	Subject309	Subject310	Subject330	Subject331	Subject332	Subject333	Subject334	Subject335	Subject337	Subject349	Subject350
10.47	295.03	168.13	183.90	256.12	262.33	260.20	269.06	248.20	202.97	328.62	228.73	266.50
Subject351	Subject352	Subject369	Subject370	Subject371	Subject372							
242.99	290.32	258.93	244.60	247.88	270.78							

What's the fuss? Using a no-pooling model



```
> mean(model$coefficients[2:length(model$coefficients)])  
[1] 251.4051
```



```
Call:
lm(formula = Reaction ~ Days * Subject - 1, data = sleepstudy)
```

Coefficients:

Days	Subject308	Subject309	Subject310	Subject330	Subject331	Subject332	Subject333	Subject334	Subject335
21.765	244.193	205.055	203.484	289.685	285.739	264.252	275.019	240.163	263.035
Subject337	Subject349	Subject350	Subject351	Subject352	Subject369	Subject370	Subject371	Subject372	Days:Subject309
290.104	215.112	225.835	261.147	276.372	254.968	210.449	253.636	267.045	-19.503
Days:Subject310	Days:Subject330	Days:Subject331	Days:Subject332	Days:Subject333	Days:Subject334	Days:Subject335	Days:Subject337	Days:Subject349	Days:Subject350
-15.650	-18.757	-16.499	-12.198	-12.623	-9.512	-24.646	-2.739	-8.271	-2.261
Days:Subject351	Days:Subject352	Days:Subject369	Days:Subject370	Days:Subject371	Days:Subject372				
-15.331	-8.198	-10.417	-3.709	-12.576	-10.467				

Linear mixed model fit by REML ['lmerMod']
Formula: Reaction ~ Days + (Days | Subject)

Data: sleepstudy
REML criterion at convergence: 1743.628
Random effects:
Groups Name Std.Dev. Corr
Subject (Intercept) 24.741
Days 5.922 0.07
Residual 25.592

Number of obs: 180, groups: Subject, 18
Fixed Effects:
(Intercept) Days
251.41 10.47

```
> sigma(lm(Reaction ~ Days * Subject - 1, data=sleepstudy))  
[1] 25.59182
```

So what's the fuss?

It flouts the assumption of independent observations!

We only had 18 subjects

Call:

```
lm(formula = Reaction ~ Days, data = sleepstudy)
```

Residual standard error: 47.71 on 178 degrees of freedom

Multiple R-squared: 0.2865, Adjusted R-squared: 0.2825

F-statistic: 71.46 on 1 and 178 DF, p-value: 9.894e-15

The GLM does not handle unbalanced datasets well

```
Call:
lm(formula = Reaction ~ Days, data = sleepstudy.NA)
Coefficients:
(Intercept)      Days
    252.69       10.31
```

Reason: GLM weights each group (subject in this case), equally despite different sample sizes

```
Linear mixed model fit by REML ['lmerMod']
Formula: Reaction ~ Days + (1 | Subject)
Data: sleepstudy.NA
REML criterion at convergence: 1593.928
Random effects:
 Groups   Name      Std.Dev.
 Subject (Intercept) 36.82
 Residual              31.43
Number of obs: 160, groups: Subject, 18
Fixed Effects:
(Intercept)      Days
    248.95       10.86
```

Experiments have many repetitions



Contents lists available at [ScienceDirect](#)

Consciousness and Cognition

journal homepage: www.elsevier.com/locate/concog



Visual expectations change subjective experience without changing performance

Lau Møller Andersen^{a,b,*}, Morten Overgaard^b, Frank Tong^c

^a NatMEG, Department of Clinical Neuroscience, Karolinska Institutet, Nobels väg 9, 171 77 Stockholm, Sweden

^b Cognitive Neuroscience Research Unit (CNRU), Center of Functionally Integrative Neuroscience (CFIN), Aarhus University, Aarhus, Denmark

^c Psychology Department, Vanderbilt University, Nashville, TN, USA

Classical strategy – Aggregation

```
str(data)
'data.frame':   25044 obs. of  18 variables:
 $ trial.type   : chr  "experiment" "experiment" "experiment" "experiment" ...
 $ pas         : Factor w/ 4 levels "1","2","3","4": 1 1 1 3 3 4 1 1 1 3 ...
 $ trial        : int   0 1 2 3 4 5 6 7 8 9 ...
 $ jitter.x     : num   -0.0608 -0.4088 0.0546 -0.4474 -0.2459 ...
 $ jitter.y     : num    0.1038 0.0166 0.4703 -0.4389 -0.4155 ...
 $ odd.digit    : Factor w/ 4 levels "3","5","7","9": 3 3 2 3 3 3 1 3 2 3 ...
 $ target.contrast: num    0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1 ...
 $ target.frames : int    3 2 1 5 6 4 3 1 2 6 ...
 $ cue          : Factor w/ 36 levels "0","1","2","3",...: 1 1 1 1 1 1 1 1 1 1 ...
 $ task         : chr    "quadruplet" "quadruplet" "quadruplet" "quadruplet" ...
 $ target.type  : chr    "odd" "odd" "even" "even" ...
 $ rt.subj      : num    1.007 0.879 0.735 1.135 1.055 ...
 $ rt.obj       : num    0.606 0.69 1.029 0.695 0.849 ...
 $ even.digit   : Factor w/ 4 levels "2","4","6","8": 3 3 4 2 3 1 2 1 4 2 ...
 $ seed         : int   93764 93764 93764 93764 93764 93764 93764 93764 93764 93764 ...
 $ obj.resp     : chr    "o" "e" "e" "e" ...
 $ subject      : Factor w/ 29 levels "1","2","3","4",...: 1 1 1 1 1 1 1 1 1 1 ...
 $ correct      : logi   TRUE FALSE TRUE TRUE TRUE TRUE TRUE ...
```

```
aggregated.data <- aggregate(rt.obj ~ subject + correct, data=data, mean)
```

*Why is
the
mean
used?*

Aggregated data

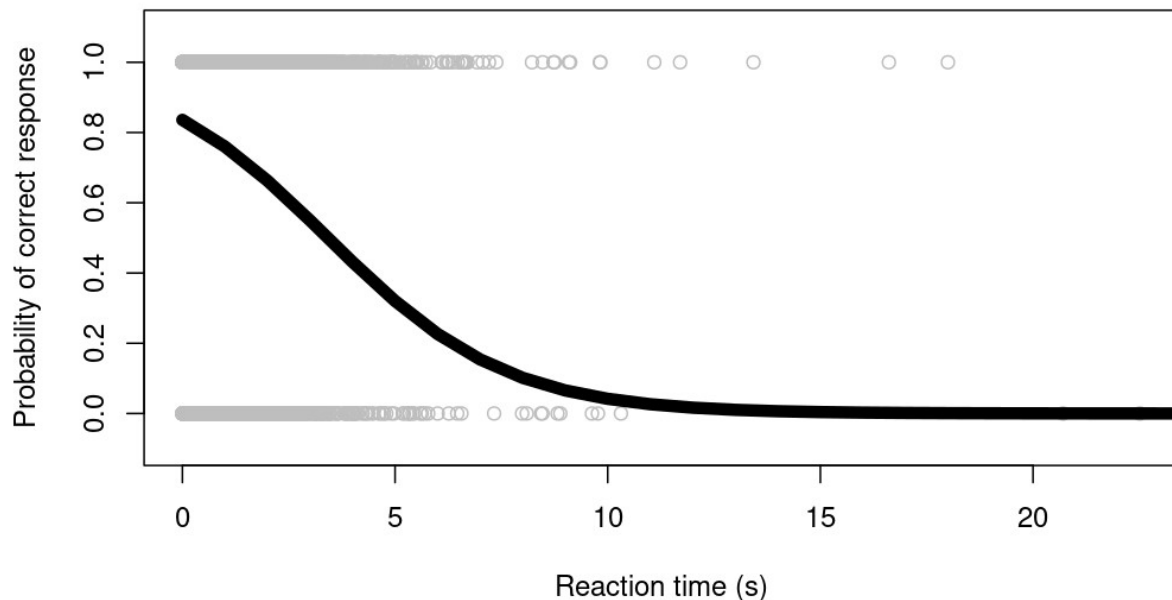
```
> str(aggregated.data)
'data.frame':  58 obs. of  3 variables:
 $ subject: Factor w/ 29 levels "1","2","3","4",...: 1 2 3 4 5 6 7 8 9 10 ...
 $ correct: logi  FALSE FALSE FALSE FALSE FALSE FALSE ...
 $ rt.obj  : num  0.673 0.586 0.882 0.785 1.178 ...
```

```
Call:
lm(formula = rt.obj ~ correct, data = aggregated.data)
Coefficients:
(Intercept)  correctTRUE
   0.9338      -0.1305
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: rt.obj ~ correct + (1 | subject)
Data: data
REML criterion at convergence: 51330.79
Random effects:
 Groups   Name      Std.Dev.
 subject (Intercept) 0.3046
Residual                  0.6721
Number of obs: 25044, groups:  subject, 29
Fixed Effects:
(Intercept)  correctTRUE
   0.9049      -0.1042
```

What about responses that are not continuous?

Logistic regression with subject-level intercepts and slopes

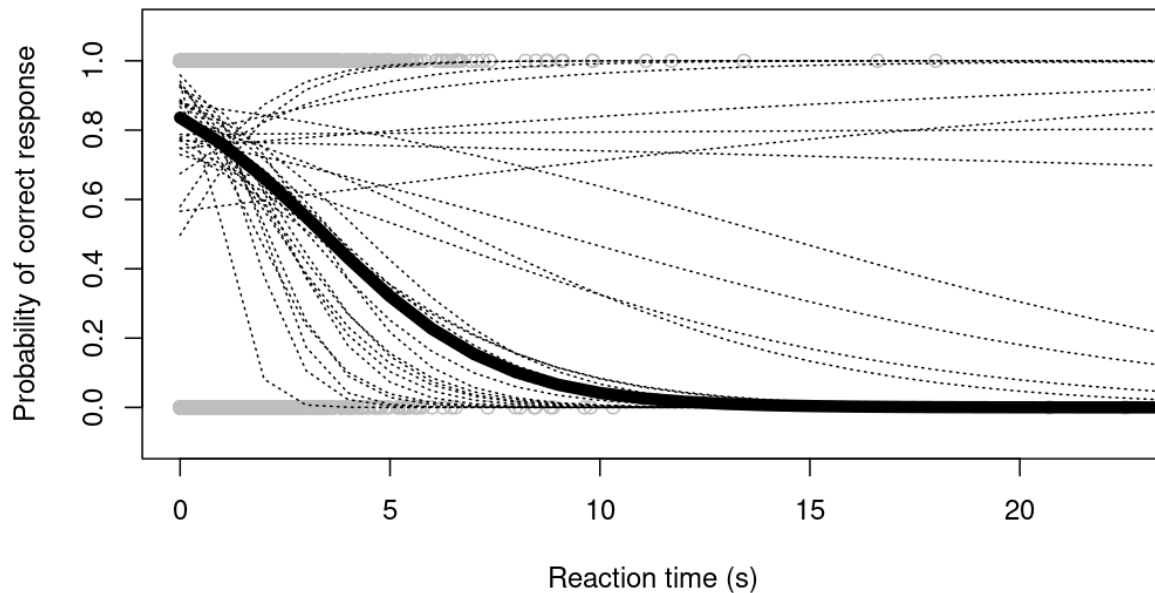


Generalized linear mixed model fit by maximum likelihood (Laplace Approximation) ['glmerMod']
Family: binomial (logit)
Formula: correct ~ rt.obj + (rt.obj | subject)

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The average may mask considerable differences

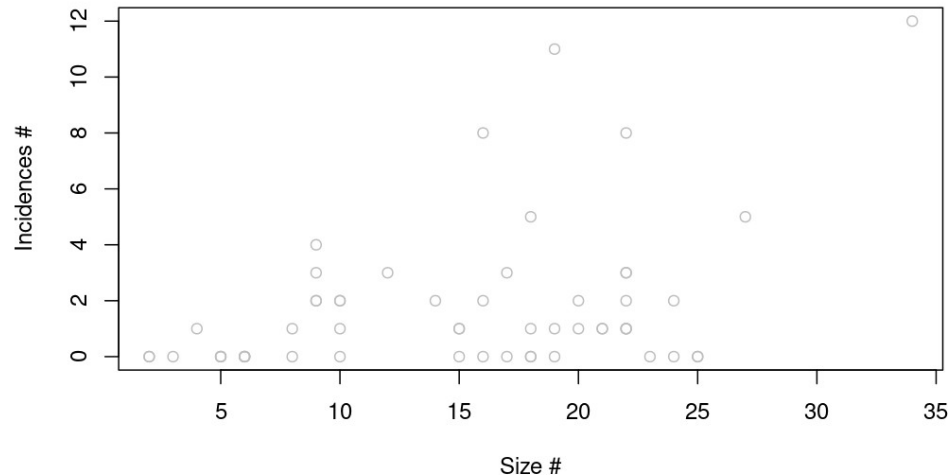
Logistic regression with subject-level intercepts and slopes



What about responses that are not continuous?

```
> print(str(cbpp))  
'data.frame': 56 obs. of 4 variables:  
 $ herd      : Factor w/ 15 levels "1","2","3","4",...: 1 1 1 1 2 2 2 3 3 3 ...  
 $ incidence: num 2 3 4 0 3 1 1 8 2 0 ...  
 $ size      : num 14 12 9 5 22 18 21 22 16 16 ...  
 $ period    : Factor w/ 4 levels "1","2","3","4": 1 2 3 4 1 2 3 1 2 3 ...
```

Poisson regression with herd- and period-level intercepts



What about responses that are not continuous?

Generalized linear mixed model fit by maximum likelihood (Laplace Approximation) ['glmerMod']

Family: poisson (log)

Formula: incidence ~ size + (1 | herd) + (1 | period)

Data: cbpp

	AIC	BIC	logLik	deviance	df.resid
	201.2789	209.3804	-96.6395	193.2789	52

Random effects:

Groups Name	Std.Dev.
-------------	----------

herd (Intercept)	0.4953
------------------	--------

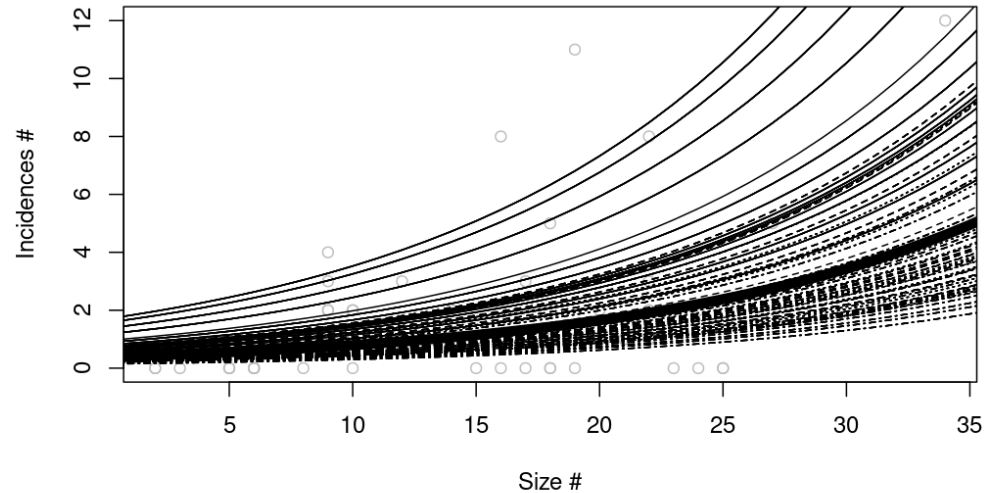
period (Intercept)	0.5137
--------------------	--------

Number of obs: 56, groups: herd, 15; period, 4

Fixed Effects:

(Intercept)	size
-0.96037	0.07312

Poisson regression with herd- and period-level intercepts



What is common to all approaches covered so far is that they *maximise* likelihood, $P(O \mid M)$, of the observations, O , given the model, M , that you are fitting

Interim summary

- Multilevel modelling handles unbalanced data well
- Multilevel modelling handles non-normal data well
- Multilevel modelling allows one to estimate effects on all levels, e.g. *subject* and *group*
- Multilevel modelling allows for using all the data, not just aggregates, e.g. *mean*, *median* or *mode*

3) Motivation for machine learning

Machine learning

Half-way into the course, we are going to turn away from *estimation* towards *prediction*

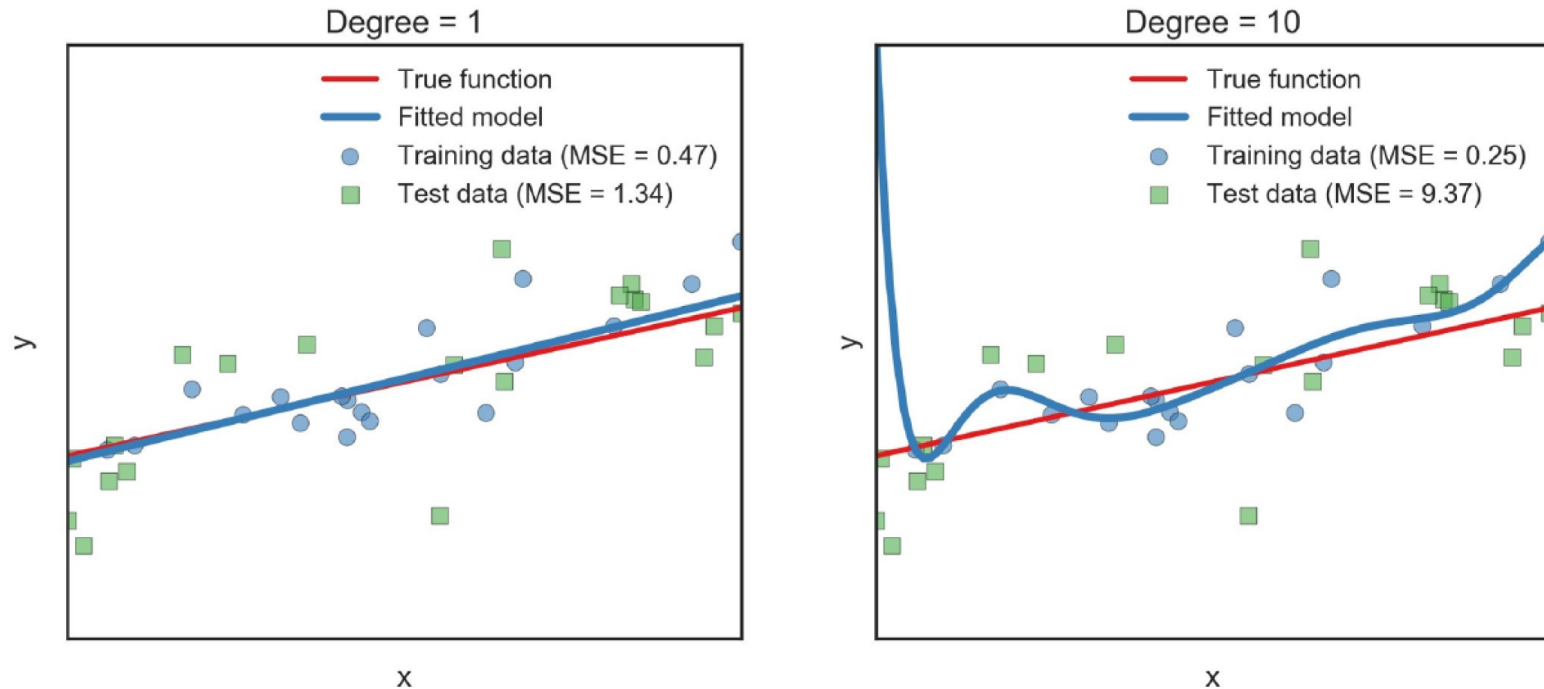
but most of the underlying mechanics are the same, as we are focusing on linear and logistic regression

The spectre of overfitting

“The tendency for statistical models to mistakenly fit sample-specific noise as if it were signal is commonly referred to as overfitting.”

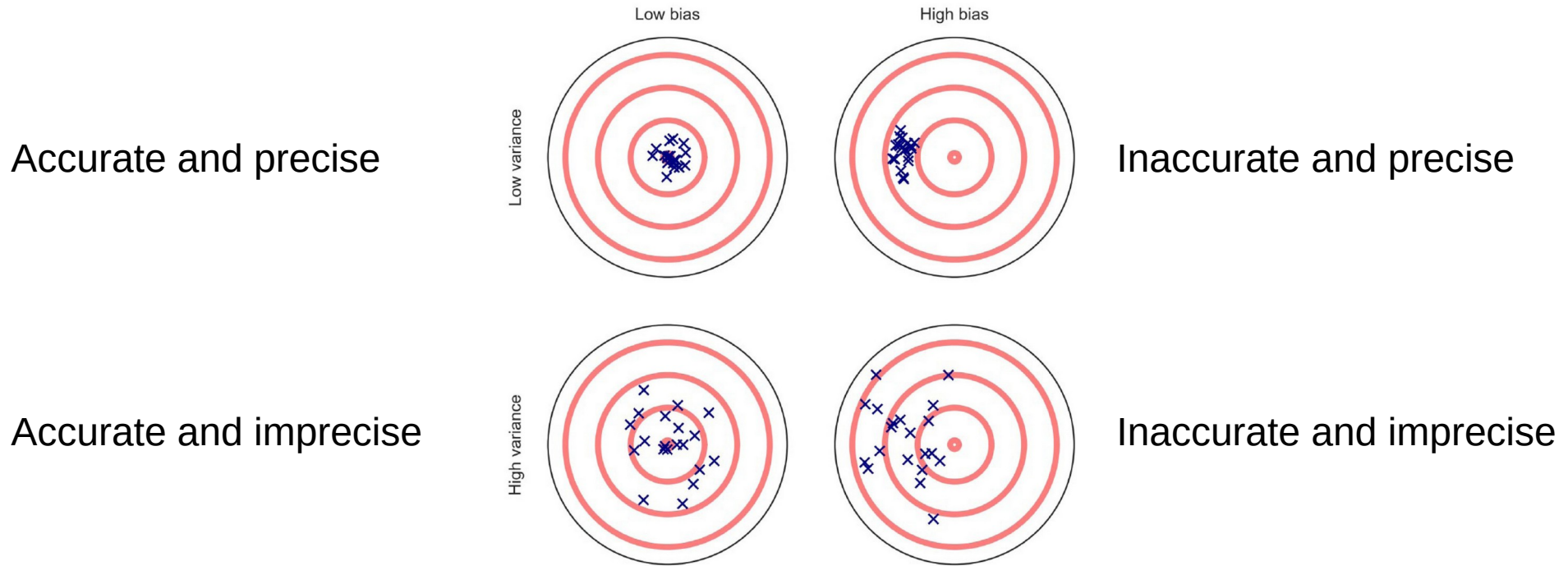
Yarkoni T, Westfall J (2017) Choosing Prediction Over Explanation in Psychology: Lessons From Machine Learning. *Perspect Psychol Sci* 12:1100–1122.
<https://doi.org/10.1177/1745691617693393>

The spectre of overfitting



Yarkoni T, Westfall J (2017) Choosing Prediction Over Explanation in Psychology: Lessons From Machine Learning. *Perspect Psychol Sci* 12:1100–1122.
<https://doi.org/10.1177/1745691617693393>

Bias-variance decomposition of error

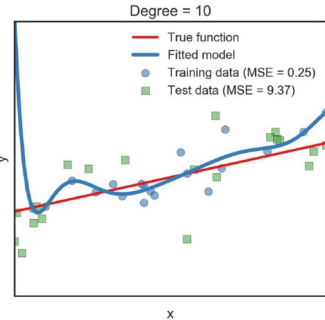


Yarkoni T, Westfall J (2017) Choosing Prediction Over Explanation in Psychology: Lessons From Machine Learning. *Perspect Psychol Sci* 12:1100–1122.
<https://doi.org/10.1177/1745691617693393>

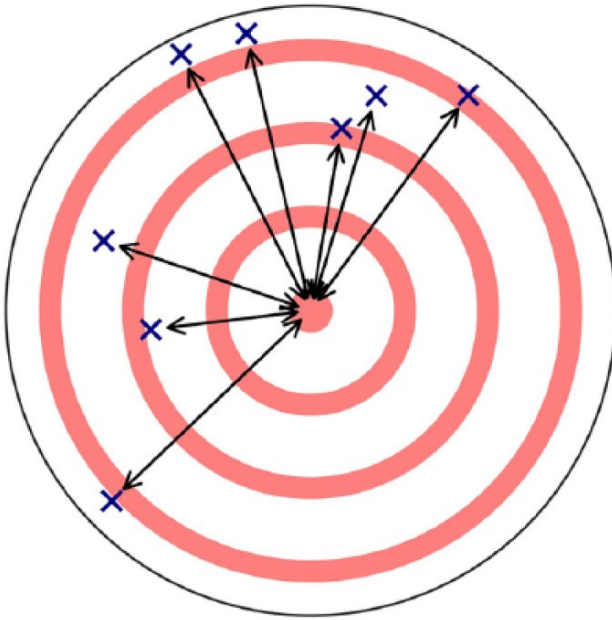
Wilfully introducing bias

WHEN WE DON'T KNOW THE TRUE FUNCTION

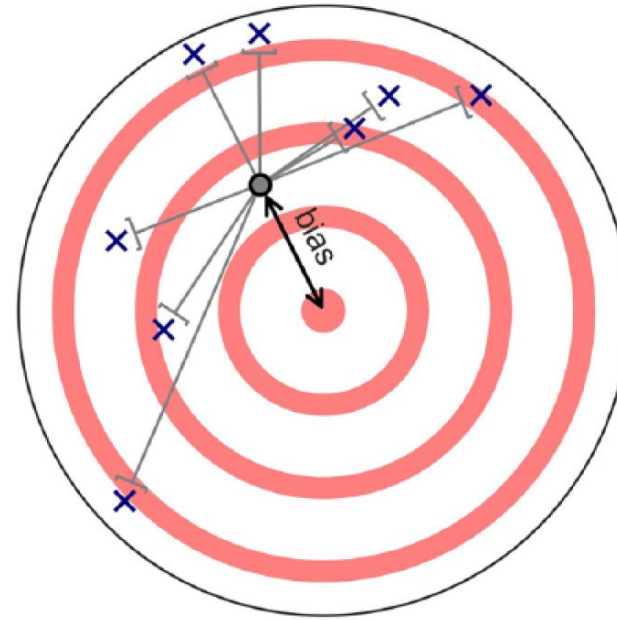
May not generalise well – due to overfitting



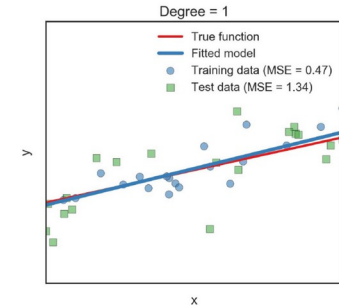
Sum of squared errors



Bias-variance decomposition



May generalise better – due to not fitting noise



Yarkoni T, Westfall J (2017) Choosing Prediction Over Explanation in Psychology: Lessons From Machine Learning. *Perspect Psychol Sci* 12:1100–1122.
<https://doi.org/10.1177/1745691617693393>

Connection between machine learning and multilevel modelling

$$l_{ML}(\hat{\beta}, \hat{u}, \hat{\sigma}^2 | y) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \log |G| - \frac{1}{2} \log |G^{-1} + Z^T Z| - \frac{n}{2} \log\left(\frac{r^2}{n}\right) - \frac{n}{2}$$

“[...] multilevel modeling approaches to analyzing clustered data [...] improve on ordinary least squares (OLS) approaches to estimating individual cluster effects by deliberately biasing (through “shrinking” or “pooling”) the cluster estimates toward the estimated population average”

Yarkoni T, Westfall J (2017) Choosing Prediction Over Explanation in Psychology: Lessons From Machine Learning. *Perspect Psychol Sci* 12:1100–1122.
<https://doi.org/10.1177/1745691617693393>

hint: “ordinary least squares”: think
methods 2

Surprising insight

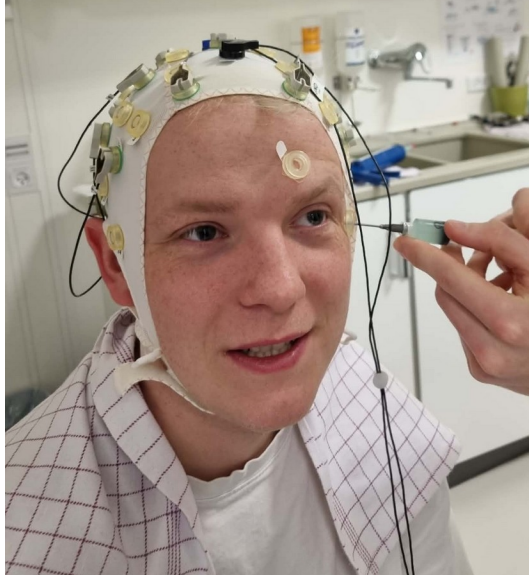
Introducing bias can improve prediction

Interim summary

- Machine learning focuses on *prediction* over *estimation*
- Machine learning includes an estimation phase, fitting the training data, and a prediction phase, seeing how well the fit predicts test data
- In fitting the training data, introducing a bias may provide better prediction of the test data
 - Bias can be introduced by using what is called regularisation
- Multilevel modelling also introduces a bias on the second level, e.g. the subject level
 - This bias shrinks the second level coefficients, away from the solution that explains the most variance

4) Presenting the instructor and myself

Instructor and lecturer

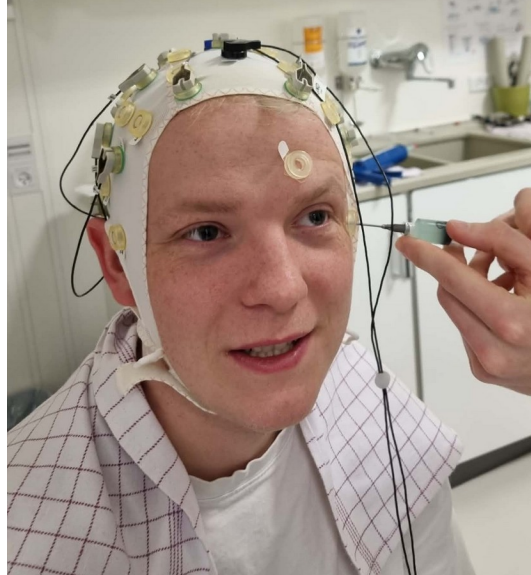


Søren Sloth
BA; 5th semester student



Lau Møller Andersen
laumollerandersen.org

Instructor and lecturer



Søren Sloth
BA; 5th semester student



Lau Møller Andersen
laumollerandersen.org

Learning your names

You

- My aim:
 - Every time I address you, I will do it with your name
 - Some times I will be wrong, and that might cause some humiliation both ways
 - Please correct me when I am wrong

Me as a lecturer

Me asking questions

- I will every now and then pick a student at random to answer questions
- I want to emphasise that I am only picking students that I judge are paying attention
 - i.e.: I am not trying to humiliate students not paying attention
- It is completely okay to say “I do not have an answer”; then I’ll just pass it on to someone else
- I will upload slides *after* the lecture

You asking questions

- Does such a thing as a stupid question exist?
- I don't care if they do or not
- You ask a question because you want to know ...
- ... and I answer because I want to tell you

Languages



<https://cran.r-project.org/mirrors.html>

Multilevel models



My recommendation:

<https://docs.conda.io/en/latest/miniconda.html>

Machine learning

Keeping up to date

The screenshot shows a GitHub repository interface. At the top, there's a navigation bar with 'main' branch selected, '1 Branch', and '0 Tags'. A search bar says 'Go to file'. There are buttons for 'Add file' and 'Code'. The repository name 'Methods-3' is visible. Below the navigation bar, a commit by 'ualsbombe' is shown with the message 'adding preamble'. A table lists files: 'week_37' (adding preamble), 'README.md' (added readme), and 'syllabus.pdf' (added syllabus). Below this is a 'README' section with the text: 'This is the README for the Methods 3 course taught at the cognitive science programme at Aarhus University during the autumn of 2025'. On the right sidebar, there's an 'About' section for 'Methods 3 Autumn 25', a 'Releases' section stating 'No releases published' with a link to 'Create a new release', and a 'Packages' section stating 'No packages published' with a link to 'Publish your first package'.

main 1 Branch 0 Tags

Go to file

Add file Code

ualsbombe adding preamble 162733e · yesterday 4 Commits

week_37	adding preamble	yesterday
README.md	added readme	yesterday
syllabus.pdf	added syllabus	yesterday

README

This is the README for the Methods 3 course taught at the cognitive science programme at Aarhus University during the autumn of 2025

About

For Methods 3 Autumn 25

Readme

Activity

Custom properties

0 stars

0 watching

0 forks

Report repository

Releases

No releases published

[Create a new release](#)

Packages

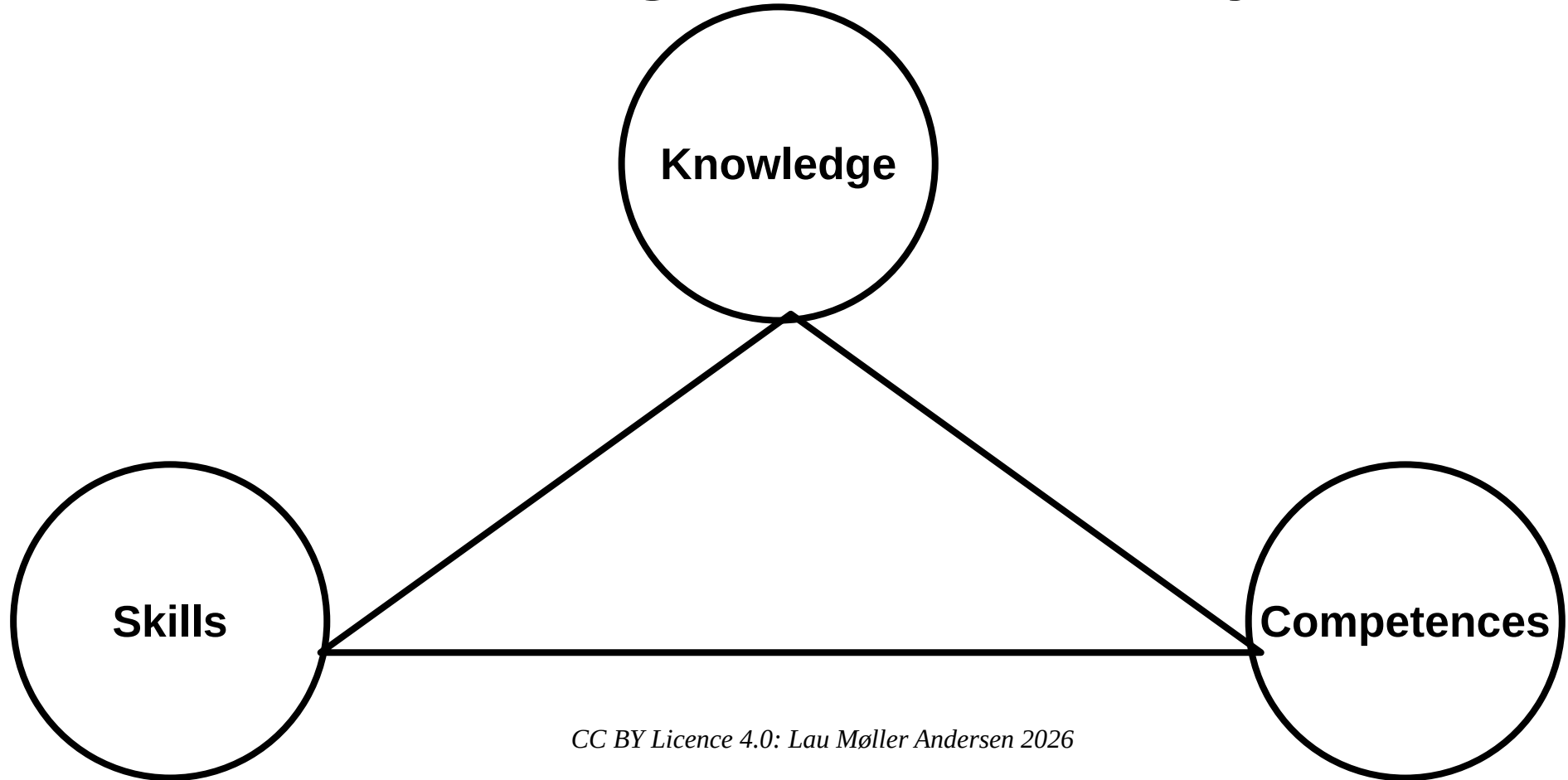
No packages published

[Publish your first package](#)

<https://github.com/Methods-3/Methods-3-F26>

5) Academic regulations and exam

Academic regulations – objectives



Academic regulations – objectives

Knowledge:

- demonstrate understanding of statistical techniques relying on the Generalised Linear Model
- demonstrate understanding of hierarchical modeling methods
- demonstrate understanding of basic machine learning concepts.

Academic regulations – KNOWLEDGE

You should thus be able to answer questions like:

What is logistic regression?

*What are some naturally
occurring hierarchies within
research data?*

What is a multilevel model?

What is cross-validation?

Academic regulations – objectives

Skills:

- build and evaluate models of hierarchically structured data
- integrate machine learning procedures in data analysis
- communicate analysis processes, results and interpretation.

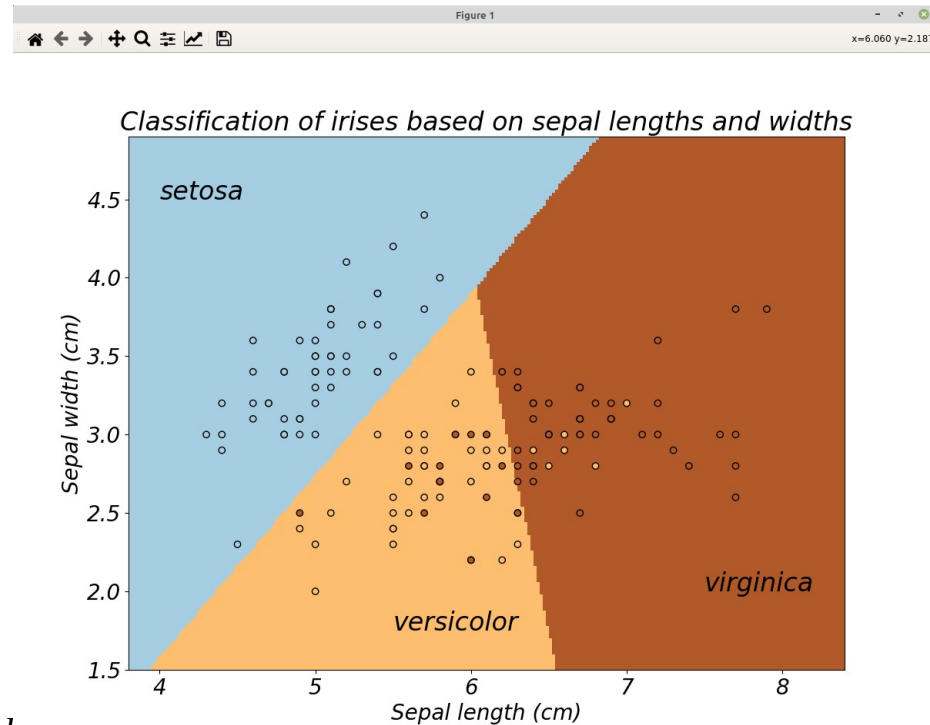
Academic regulations – SKILLS

*build and evaluate models of
hierarchically structured data*

```
Linear mixed model fit by REML ['lmerMod']  
Formula: Height ~ Gender + (1 | Family)  
Data: height
```

Academic regulations – SKILLS

*integrate machine learning procedures
in data analysis*



Academic regulations – SKILLS

*communicate analysis processes,
results and interpretation*

(Andersen et
al., 2019)

Mixed model analyses (McCulloch & Neuhaus, 2005) were applied to investigate how top-down expectations (No. of Possible Targets) affected subjective experience and objective performance. We performed model comparisons between models that did or did not include the relevant fixed effects and interactions to find the best compromise between an explanatory and a parsimonious model. This was done using the log-likelihood ratio between two models because this ratio approximates a chi-square distribution. A chi-square test can thus be used to assess whether two models differ significantly, where the test statistic is the log-likelihood-ratio and the degrees of freedom is the difference in free parameters of the two models.

Academic regulations – objectives

Competences:

- independently decide on data analysis methods, given a data set and a research question
- justify decision making when pre-processing messy data for data analysis.

Difference between *skills* and *competences*:
Skills are specific
Competences are more generic

Academic regulations – COMPETENCES

*independently decide on data analysis
methods, given a data set and a
research question*

*justify decision making when pre-
processing messy data for data analysis.*

EXAMPLE QUESTION: Can native
Danish speakers tell apart soft d's (ð)
and l's if they are embedded in
English speech, e.g.
maðfunction/malfunction

EXAMPLE DATA: Dataset with
response times and discrimination
responses



Exam; portfolio

- Ongoing assignments to be solved in study groups
 - I need **knitted** assignments; e.g. *RMarkdown*, *nbconvert*, *Quarto*; hand in the knitted pdf **and** the basis for the knitting, e.g. *.Rmd*, *.qmd*, *.ipynb*
- Final portfolio:
 - Revised assignments, handed in as short reports with reproducible code in GitHub Classroom
 - Assignment 0: Using mixed effects modelling to model hierarchical data (Winter & Grawunder, 2012)
 - Assignment 1: Mixed effects modelling of accuracy (Andersen et al., 2019)
 - Assignment 2: Using logistic regression to classify subjective experience from brain data (Sandberg et al., 2016)
 - Assignment 3: Dimensionality reduction; finding the signal among the noise

Me and you

- Mutual respect
- I teach this course because I care about it; i.e. I invest **professional pride** (Danish: *faglig stolthed*) in it
- I respect that you all have different backgrounds and different ways of engaging with me (see my stance on questions)

Brave new world

WHEN MY PROFESSIONAL PRIDE TOOK A BLOW

Professional pride -> Professional cynicism (Danish: *faglede*)



Rules and recommendations

- Use them as little as possible; and ask very specific questions. Don't have it solve a whole question for you
- Don't cheat!
 - Rules: <https://studerende.au.dk/en/gai>

A powerful tool ...

- Part of my professional pride is that when people don't show mutual respect, I take action
 - we are not idiots
 - (it ain't that hard to detect; I reported several cases in 2025, which had consequences)
- But first and foremost – I want to teach you – that's why I am here
 - and I can't teach you if you use GAI to do your exam and your learning for you

Re-examination

Will be done during the summer (tip: stay on schedule, such that you can enjoy your summer)

The structure of lectures

- Tuesdays 10.00-12.00 (NB: Academic quarter)
- A mix between the general theoretical framework and a few formulas
- Code will be incorporated
- Group discussions
- Please don't hold back on your questions

The structure of a practical exercise

- Thursdays 08.00-10.00 and 10.00-12.00
- The practical exercises are led by either Søren or me.
 - An introduction to the task
 - Data and an instructions file are shared
 - Work in study groups
 - Intermittent discussion of pressing issues
- Deadline:
 - On the day of the next class (23.59). i.e. a bit more than a week.
- Feedback policy
 - Detailed written feedback will be provided for **Assignment 0 only**, with the aim of aligning expectations of the instructor and students regarding how assignments will generally be read and graded. This **has to be** handed in as a **study group assignment** – otherwise you do **NOT** get feedback. **Oral feedback** will be provided on subsequent portfolio assignments during in-class assignment preparation sessions, where students will have ample opportunity to ask the instructor questions on their assignments-in-progress.
 - Note for all assignments; feedback is only given if handed in on time

Student counsellors



Christian: Cognitive
Science



Charlotte:
Linguistics and
Cognitive Semiotics



Sofia: Scandinavian
Languages and
Literature



Pernille:
Experience
Economy and
Cultur Of Events

<https://studerende.au.dk/en/studies/subject-portals/arts/counselling/student-counsellor/nordiskeng>

The course plan

Week 37:

Lesson 0: What is it all about?

Class 0: Setting up R and Python — and recollection of the general linear model

Week 38:

Lesson 1: Multilevel linear regression

Class 1: Modelling subject level effects – and how do they differ from group level effects?

Week 39:

Lesson 2: Beyond linear regression — link functions

Class 2: What to do when the response variable is not continuous?

Week 40:

Lesson 3: Evaluating generalised linear mixed models — and the spectre of collinearity

Class 3: Code review

Week 41:

Lesson 4: Explanation versus prediction

Class 4: Summarising mixed models

Week 42:

no teaching (go fetch potatoes instead)

Week 43:

Lesson 5: Mid-way evaluation and machine learning introduction

Class 5: Getting Python running

Week 44:

Lesson 6: Linear and logistic regression revisited (machine learning)

Class 6: Categorizing responses based on informed guesses

Week 45:

Lesson 7: Model evaluation and hyperparameter tuning

Class 7: Code review

Week 46:

Lesson 8: Dimensionality reduction — principled component analysis (PCA)

Class 8: What to do with very rich data?

Week 47:

Lesson 9: Neural networks

Class 9: Code review

Week 48:

Lesson 10: Unsupervised classification

Class 10: Create video explainers of central concepts (shared resource)

Week 49:

Lesson 0 again: What was it all about? and final evaluation

Class 11: Summarising machine learning

Week 50:

Student presentations:

Class 12: Work on portfolios: Ask anything!

Next time – multilevel linear regression

- Design matrix Z
- Pooling, no pooling and partial pooling
- How multilevel models are fit
- How bias, not necessarily a bad thing, is introduced in multilevel modelling

Reading questions

- Gelman and Hill:
 - Chapter 11
 - Do the motivations, section 11.5, apply to cognitive science research as you know it?
 - Chapter 12
 - What does pooling mean?
 - What is complete pooling?
 - What is no pooling?
 - What is partial pooling?
 - How is shrinkage achieved in partial pooling?
 - What does it mean that classical regression is a special case of multilevel modelling?
 - Have a good look at equations 12.1 and 12.4
 - How does the discussion about number of groups and number of observations per group relate to cognitive science research as you know it?
- Bates et al. 2015
 - You do not need to understand all the algebra of section 3, but have a look anyway
 - What does it mean that the random variable Y in eq. 2, is conditional on the instantiation, $\mathbf{b}$, of the random variable B ?
 - How is the $\|u^2\|$ term in eq. 13 related to the bias, shrinkage, introduced in multilevel modelling? (see the machine learning summary above)?